



Implementing Lightweight Intrusion Detection System on Resource Constrained Devices

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Introduction

- IoT devices face security challenges due to resource limitations.
- Traditional IDS solutions are unsuitable for IoT environments because of their high resource demands.
- Our lightweight IDS combines signature-based and anomaly-based detection to secure IoT networks with minimal computational overhead.
- Evaluated on Raspberry Pi 4 using the CICIDS2017 dataset and real-world simulations.

Related Work

TABLE I
SUMMARY OF STUDIES ON IDS IMPLEMENTATIONS

Study	Platform	Method	Machine Learning Techniques
A. Sforzin <i>et al.</i> [7]	Raspberry Pi 2	Signature-based	None
S. Tripathi <i>et al.</i> [8]	Raspberry Pi 3	Signature-based	None
P. A. M. Hambali <i>et al.</i> [9]	Raspberry Pi 3	Signature-based	None
N. Sivanandam <i>et al.</i> [10]	Raspberry Pi	Anomaly-based	Decision Tree (DT), Random Forest (RF)
E. A. Katonová <i>et al.</i> [11]	Raspberry Pi 4 Model B	Anomaly-based	Random Forest (RF), Multi-Layer Perceptron (MLP), Support Vector Classifier (SVC), K-Nearest Neighbors (KNN), Extreme Gradient Boosting (XGB)
P. R. Agbedanu <i>et al.</i> [12]	Raspberry Pi 2	Anomaly-based	Incremental Principal Component Analysis (IPCA), Self-Adjusting Memory K-Nearest Neighbors (SAM-KNN)
R. Sumanth <i>et al.</i> [13]	Raspberry Pi 3	Anomaly-based	K-Means Clustering
G. A. Morales <i>et al.</i> [14]	Raspberry Pi 2	Anomaly-based	Support Vector Machine (SVM), Logistic Regression (LR), K-Nearest Neighbors (KNN), Random Forest (RF)
E. Ciklabakkal <i>et al.</i> [15]	Not specified	Anomaly-based	Autoencoder (AE), Self-Organizing Generative Adversarial Active Learning (SO-GAAL), Random Forest (RF), K-Means Clustering, One-Class Support Vector Machine (OCSVM), Isolation Forest (IF)

Proposed System Overview

- Real-time monitoring and detection system for IoT networks.
- Combines signature-based detection and machine learning-based anomaly detection.
- Designed for Raspberry Pi 4 Model B.

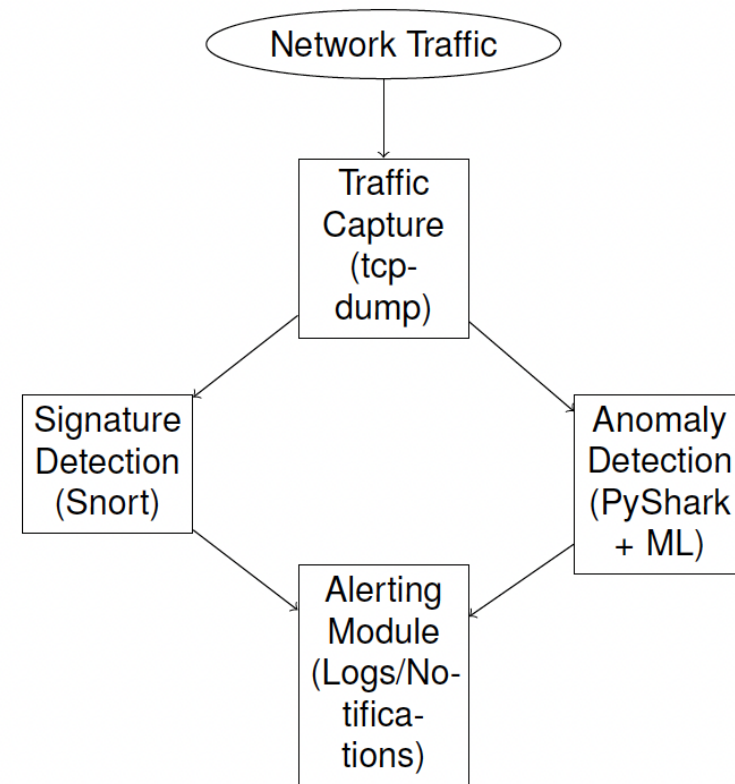


Fig. 1. Proposed IDS Architecture

System Components

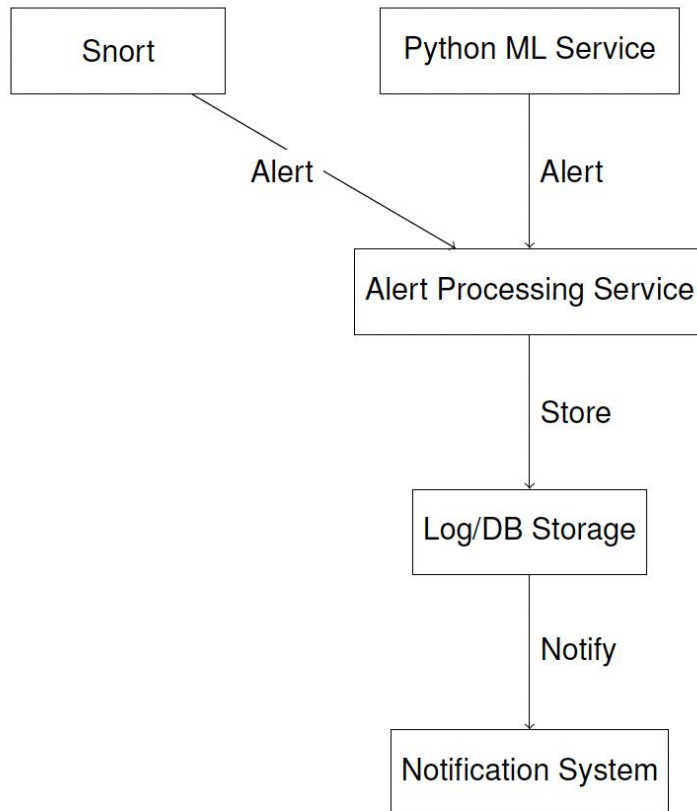


Fig. 2. Alert System Architecture

- Traffic Capture: Using tcpdump and PyShark for traffic analysis.
- Threat Detection: Combines Snort (signature-based) and machine learning models like Random Forest and K-Nearest Neighbors (KNN).
- Alerting Module: Logs and future real-time notifications.

Anomaly-Based Detection

- ML Models: Random Forest and K-Nearest Neighbors (KNN).
- Dataset: CICIDS2017.
- Features: Top 20 selected features.

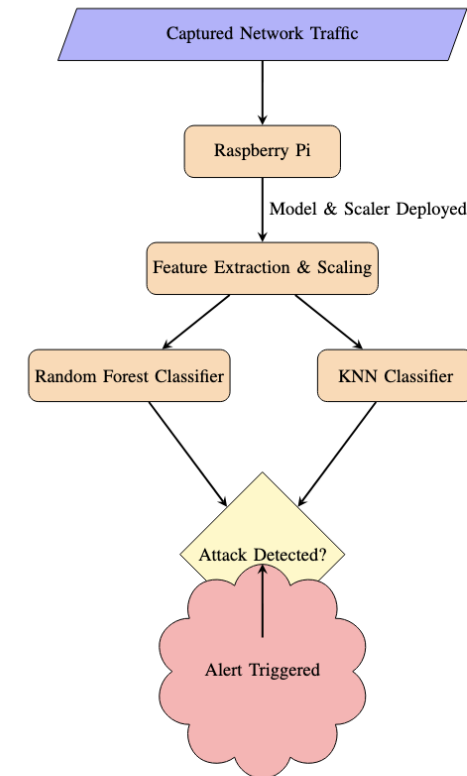


Fig. 3. Overview of Model Deployment and Attack Detection Process on Raspberry Pi

Anomaly-Based Detection Performance

- Models: Random Forest and K-Nearest Neighbors (KNN).
- Dataset: CICIDS2017, includes attacks such as:
 - Brute-force attacks (FTP-Patator, SSH-Patator)
 - Denial of Service (DoS) (Slowloris, Hulk, GoldenEye)
 - Web-based attacks (XSS, SQL injection)
 - Other threats: DDoS, port scanning, botnet activity.

Implementation and Results

TABLE II
RANDOM FOREST CLASSIFICATION REPORT

Class	Precision	Recall	F1-Score	Support
Benign Traffic	0.9906	0.9717	0.9811	557646
Attack Traffic	0.9722	0.9908	0.9814	557646
Accuracy	0.9813			
Macro Avg	0.9814	0.9813	0.9813	1115292
Weighted Avg	0.9814	0.9813	0.9813	1115292

TABLE III
RANDOM FOREST CONFUSION MATRIX

	Predicted: Benign	Predicted: Attack
Actual: Benign	541870 (TN)	15776 (FP)
Actual: Attack	5118 (FN)	552528 (TP)

TABLE IV
KNN CLASSIFICATION REPORT

Class	Precision	Recall	F1-Score	Support
Benign Traffic	0.9880	0.9732	0.9806	557646
Attack Traffic	0.9736	0.9882	0.9809	557646
Accuracy	0.9807			
Macro Avg	0.9808	0.9807	0.9807	1115292
Weighted Avg	0.9808	0.9807	0.9807	1115292

TABLE V
KNN CONFUSION MATRIX

	Predicted: Benign	Predicted: Attack
Actual: Benign	542728 (TN)	14918 (FP)
Actual: Attack	6588 (FN)	551058 (TP)

Signature-Based Detection

```
[**] [1:1000001:1] Ping detected [**]  
[Priority: 0]  
08/11-19:11:42.404627 192.168.0.45 -> 192.168.0.1  
ICMP TTL:64 TOS:0xC0 ID:37172 IpLen:20 DgmLen:106  
Type:3 Code:3 DESTINATION UNREACHABLE: PORT UNREACHABLE
```

```
[**] [1:1000002:1] SYN Flood detected [**]  
[Priority: 0]  
08/11-18:50:06.031504 192.168.0.18:42706 -> 192.168.0.45:80  
TCP TTL:64 TOS:0x0 ID:27365 IpLen:20 DgmLen:40  
*****S* Seq: 0x33B219CF Ack: 0x3924152F Win: 0x200 TcpLen: 20
```

```
[**] [1:1000007:1] SSH Brute-force attempt detected [**]  
[Priority: 0]  
08/11-19:07:28.398231 192.168.0.18:39432 -> 192.168.0.45:22  
TCP TTL:64 TOS:0x0 ID:40965 IpLen:20 DgmLen:74 DF  
***AP*** Seq: 0x7F2BD8C8 Ack: 0x10148946 Win: 0x1F6 TcpLen: 32  
TCP Options (3) => NOP NOP TS: 2691466795 3149052523
```

- Tested against Ping of Death, SYN Flood, and SSH Brute-Force.
- Snort effectively detected all tested attacks.

TABLE VI
THREAT DETECTION TEST RESULTS

Attack Type	Tool	Verification
Ping of Death	hping3	Monitored ICMP alerts
SYN Flood	hping3	Monitored SYN alerts
SSH Brute-Force	Hydra	Monitored failed login alerts

Computational Efficiency

- Snort: 54.3% CPU usage.
- Python ML service: 73.3% CPU usage.
- System maintained stability under high traffic conditions.

TABLE VII
CPU USAGE DURING LOAD TEST

Process	CPU Usage (%)
snort (PID: 64034)	54.3
python3 (PID: 65489)	73.3

Conclusion

- IoT growth demands efficient and accurate security solutions.
- Our proposed lightweight IDS for Raspberry Pi combines signature-based detection (Snort) and machine learning models for anomaly detection.
- Experimental results show high accuracy and efficient processing in detecting attacks

Future Work

- Distributed IDS using a Raspberry Pi cluster.
- Extend to ESP32 for lightweight IoT nodes.
- Implement parallel detection and load balancing.
- Develop self-healing capabilities.



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